

PRATEEK SINGH

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EDUCATION

University of Washington

Sep 2025 – Mar 2027

Master of Science in Computer Science and Software Engineering

GPA: 3.8/4.0

- **Courses:** Distributed Computing, Parallel Programming in Grid and Cloud, Advanced Algorithms, Database Systems Design, Advanced Machine Learning, Applied Mathematics and Scientific Computing

University of Mumbai

Jun 2019 – May 2023

Bachelor of Engineering in Information Technology

GPA: 8.11/10

SKILLS

Languages: C++, C, JavaScript, Python, Java, SQL, Swift

Frameworks: Spring Boot, Django, Flask, React, Node.js, TensorFlow, PyTorch

Cloud & DevOps: AWS (Lambda, EC2, ECS, S3), Docker, Kubernetes, Jenkins, Terraform, CI/CD

Databases: PostgreSQL, MongoDB, Cassandra, Redis, MySQL, DynamoDB

Tools: Git, GitHub, JIRA, Linux/Unix, REST APIs

EXPERIENCE

Tata Communications – Software Development Engineer

Jun 2023 – Jun 2025

- Developed a scientist paper-read assistant using an MCP-based architecture to help users retrieve, interpret, and synthesize academic literature.
- Built capabilities for semantic paper search, citation traversal, paper summarization, and question answering over scholarly content and metadata.
- Enhanced system quality with embeddings-based retrieval, reranking, chunk-level document understanding, and citation-grounded answer generation.
- Supported researcher productivity use cases including literature survey, related-paper discovery, and multi-document insight extraction.

Effcode Technologies – Software Engineer Intern

Jun 2022 – Dec 2022

- Developed and deployed Node.js payment APIs with 30% higher transaction reliability.
- Redesigned legacy Windows applications as single page apps in React and Python.
- Engineered a network architecture for hosting web applications on remote client machines using Nginx.
- Increased test coverage to 85% using Jest and integration tests.

Remote

ACADEMIC/RESEARCH PROJECTS

Music Streaming Platform | React, Node.js, Redis, WebSocket

- Developed distributed app supporting 100+ concurrent users with real-time synchronized playback.
- Implemented CRDT-based state management (Yjs) ensuring conflict-free synchronization.
- Utilized Redis caching for frequently accessed tracks, reducing latency and database load.

AWS Lambda Cost Optimizer | Python, AWS Lambda

- Built self-service analyzer detecting inefficient Lambda configurations, reducing monthly cloud cost by 28%.
- Open-sourced the tool with comprehensive documentation.

Women's Fashion and Beauty Shopping Agent

Jun 2021 – May 2022

- Developed an AI-powered shopping assistant for women's fashion and beauty that supported conversational product discovery across clothing, skincare, makeup, and accessories.
- Integrated MCP-compatible tools for catalog retrieval, review analysis, recommendation ranking, and personalized bundle generation based on user preferences and constraints.
- Implemented agent-driven flows for outfit coordination, skincare recommendation, product comparison, and cart building to improve shopping guidance beyond standard search and filtering.
- Kept a human in the loop by limiting the agent to shortlisting items and adding selected products to cart, while leaving the final checkout and purchase decision fully manual.

CERTIFICATIONS

- Generative AI with LLMs (DeepLearning.AI, IBM)
- Data Structures and Algorithms (CodeHelp)
- AWS Solutions Architect (in progress)